

# WEBRUM-05: Error Analysis

---

**Series:** WEBRUM | **Notebook:** 5 of 8 | **Created:** March 2026 | **Last Updated:** 03/12/2026

## Overview

---

JavaScript errors are among the most impactful issues affecting web application user experience. A single unhandled exception can break page functionality, prevent form submissions, or cause entire features to stop working. Dynatrace RUM captures JavaScript errors, XHR/fetch errors, and custom errors in real-time, enabling you to quantify error impact on real users.

This notebook covers error types, error grouping, impact analysis (how many sessions/users are affected), rage click detection, and error-to-session correlation using DQL.

---

## Table of Contents

---

1. [Error Types in RUM](#)
  2. [Error Volume and Trends](#)
  3. [Error Grouping and Top Errors](#)
  4. [Error Impact Analysis](#)
  5. [XHR and Fetch Errors](#)
  6. [Rage Click Detection](#)
  7. [Error-to-Session Correlation](#)
  8. [Summary and Next Steps](#)
-

# Prerequisites

| Requirement                  | Details                                    |
|------------------------------|--------------------------------------------|
| <b>Dynatrace Environment</b> | SaaS with Grail enabled                    |
| <b>RUM Enabled</b>           | Web applications with error capture active |
| <b>Permissions</b>           | <code>storage:events:read</code>           |
| <b>Previous Notebook</b>     | WEBRUM-01: RUM Fundamentals                |

## 1. Error Types in RUM

Dynatrace captures several categories of browser-side errors:

| Error Type              | Description                                                         | Example                                                       |
|-------------------------|---------------------------------------------------------------------|---------------------------------------------------------------|
| <b>JavaScript error</b> | Unhandled exception or runtime error                                | <code>TypeError: Cannot read property 'x' of undefined</code> |
| <b>XHR error</b>        | Failed XHR/fetch request (HTTP 4xx/5xx or network error)            | <code>HTTP 500 on POST /api/checkout</code>                   |
| <b>Custom error</b>     | Errors reported via the RUM API ( <code>dtum.reportError()</code> ) | <code>Payment validation failed</code>                        |
| <b>CSP violation</b>    | Content Security Policy blocking a resource                         | <code>Refused to load script from 'http://evil.com'</code>    |
| <b>Resource error</b>   | Failed resource loading (images, scripts, stylesheets)              | <code>Failed to load /assets/app.js</code>                    |

## Error Data Model

| Field                      | Description                                 |
|----------------------------|---------------------------------------------|
| <code>error.message</code> | Error message text                          |
| <code>error.type</code>    | Error classification                        |
| <code>error.source</code>  | Source file and line number                 |
| <code>action.name</code>   | User action during which the error occurred |
| <code>session.id</code>    | Session containing the error                |
| <code>app.name</code>      | Application name                            |

```
// Recent RUM errors – explore the data structure
fetch dt.rum.error, from:-1h
| fieldsKeep timestamp, error.message, error.type, error.source, action.name,
app.name, session.id
| sort timestamp desc
| limit 20
```

## 2. Error Volume and Trends

Start by understanding overall error volume and how it changes over time.

```
// Error count by type over last 24 hours
fetch dt.rum.error, from:-24h
| summarize error_count = count(), by:{error.type}
| sort error_count desc
```

```
// Error trend over 24 hours – hourly bucketed by error type
fetch dt.rum.error, from:-24h
| makeTimeseries error_count = count(), interval:1h, by:{error.type}
```

```
// Error volume by application – which apps have the most errors?
fetch dt.rum.error, from:-24h
| summarize error_count = count(),
  unique_errors = countDistinct(error.message),
  affected_sessions = countDistinct(session.id),
  by:{app.name}
| sort error_count desc
```

### 3. Error Grouping and Top Errors

---

Error messages often contain variable data (stack traces, URLs, IDs). Grouping by error message helps identify the most frequent issues.

```
// Top 15 errors by frequency – the most common error messages
fetch dt.rum.error, from:-24h
| summarize error_count = count(),
  affected_sessions = countDistinct(session.id),
  by:{error.message, error.type}
| sort error_count desc
| limit 15
```

```
// Errors by page – which pages generate the most errors?
fetch dt.rum.error, from:-24h
| filter isNotNull(action.name)
| summarize error_count = count(),
  unique_errors = countDistinct(error.message),
  by:{action.name}
| sort error_count desc
| limit 10
```

### 4. Error Impact Analysis

---

Not all errors are equal. An error affecting 1 session is different from one affecting 10,000. Impact analysis quantifies the blast radius of each error.

```
// Error impact – errors ranked by number of affected sessions
fetch dt.rum.error, from:-24h
| summarize total_occurrences = count(),
    affected_sessions = countDistinct(session.id),
    by:{error.message}
| sort affected_sessions desc
| limit 10
```

```
// Error rate per application – percentage of sessions with errors
fetch dt.rum.user_session, from:-24h
| filter user.type == "REAL_USER"
| summarize total_sessions = count(),
    error_sessions = countIf(error.count > 0),
    by:{app.name}
| fieldsAdd error_rate_pct = round(toDouble(error_sessions) /
toDouble(total_sessions) * 100.0, decimals: 2)
| sort error_rate_pct desc
```

**Tip:** Focus on errors with high session impact rather than high occurrence count. A single user hitting a retry loop can generate thousands of occurrences from one session.

## 5. XHR and Fetch Errors

---

XHR/fetch errors indicate backend API failures impacting the frontend. These are particularly critical in SPAs where the UI depends entirely on API responses.

```
// XHR errors by action – identify failing API calls
fetch dt.rum.error, from:-24h
| filter error.type == "XHR_ERROR" or error.type == "FETCH_ERROR"
| summarize error_count = count(),
    affected_sessions = countDistinct(session.id),
    by:{error.message, action.name}
| sort error_count desc
| limit 15
```

```
// XHR error trend – are backend errors increasing?
fetch dt.rum.error, from:-24h
| filter error.type == "XHR_ERROR" or error.type == "FETCH_ERROR"
| makeTimeseries error_count = count(), interval:1h
```

## 6. Rage Click Detection

---

A **rage click** occurs when a user rapidly clicks the same element multiple times — a strong signal of frustration. This usually indicates:

- A button that appears clickable but is not responding
- A form submission that is silently failing
- A UI element that is loading too slowly
- A dead link or broken navigation element

Dynatrace captures rage clicks as user action properties. They can also be detected through rapid action sequences:

```
// Sessions with rage clicks – identify frustrated users
fetch dt.rum.user_action, from:-24h
| filter isNotNull(rage.click) and rage.click == true
| summarize rage_count = count(),
  by:{action.name, app.name}
| sort rage_count desc
| limit 10
```

```
// Rage click trend over 7 days – is frustration increasing?
fetch dt.rum.user_action, from:-7d
| filter isNotNull(rage.click) and rage.click == true
| makeTimeseries rage_count = count(), interval:1d
```

## 7. Error-to-Session Correlation

---

Understanding how errors correlate with session outcomes (bounce, low engagement, failed conversion) helps prioritize fixes.

```
// Compare sessions with and without errors – engagement impact
fetch dt.rum.user_session, from:-24h
| filter user.type == "REAL_USER"
| fieldsAdd has_errors = if(error.count > 0, "With Errors", else: "No Errors")
| summarize session_count = count(),
    avg_actions = avg(action.count),
    avg_duration_sec = avg(toDouble(duration) / 1000000000.0),
    bounce_rate = countIf(action.count == 1),
    by:{has_errors}
| fieldsAdd bounce_pct = round(toDouble(bounce_rate) /
toDouble(session_count) * 100.0, decimals: 1)
```

```
// Errors by browser – are certain browsers more error-prone?
fetch dt.rum.user_session, from:-24h
| filter user.type == "REAL_USER"
| filter isNotNull(browser.family)
| summarize total = count(),
    with_errors = countIf(error.count > 0),
    by:{browser.family}
| fieldsAdd error_rate = round(toDouble(with_errors) / toDouble(total) *
100.0, decimals: 1)
| filter total > 10
| sort error_rate desc
| limit 10
```

## 8. Summary and Next Steps

---

In this notebook, we covered:

- **Error types** — JavaScript errors, XHR errors, custom errors, resource errors
- **Error volume and trends** — Tracking error rates over time
- **Error grouping** — Identifying the most frequent error messages
- **Impact analysis** — Measuring how many sessions each error affects
- **XHR/fetch errors** — Backend API failures visible from the frontend
- **Rage clicks** — Detecting user frustration through rapid repeated clicks
- **Error-session correlation** — How errors impact engagement and bounce rates

## Next Steps

- **WEBRUM-06: Performance Analysis** — Page load waterfall and performance bottleneck identification
- **WEBRUM-07: Session Replay** — Visually investigate error-impacted sessions

## References

- [JavaScript Error Tracking](#)
- [Rage Click Detection](#)

---

*This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.*